

Experience	Avionics Research Engineer – Georgia Tech Research Institute Aug 2025 to Present – Atlanta, GA - Developing C++ software for collaborative autonomous UAV swarms in a component-based software engineering environment. - Developed a containerized motion planning service with hybrid planning using graph, sampling, and optimization methods. - Integrating, developing, and benchmarking sampling-based motion planning algorithms over constraint manifolds – leveraging The Open Motion Planning Library (OMPL). - Designed and developed a world model service for a collaborative-decentralized swarm. - Designed and developed a multi-object tracking (MOT) library for decentralized robot swarms using strategy design pattern - Implemented prediction and data association algorithms leveraging consensus algorithms - Wrote data association algorithms for multi-object tracking (MOT) and vehicle-to-vehicle track consensus. - Wrote distributed state-estimation data-association algorithms for distributed multi-sensor tracking. - Software systems integration		
	Robotics Researcher / M.S. Thesis – Virginia Tech Aug 2023 to May 2025 – Blacksburg, VA - Built a 6-DOF robotic arm and an accompanying ROS2–Gazebo simulation environment for training, evaluating, and deploying custom control algorithms. - Integrated object detection (YOLOv8), natural language processing, symbolic reasoning, and a DDPG-based reinforcement learning policy for robotic manipulation tasks. - Aided professors in teaching fundamental concepts in linear systems theory and digital signal processing, including Laplace Transforms, Z-Transforms, system stability, and FIR & IIR filter design. - Assisted with hands-on projects to illustrate and integrate analog and digital filter design and application on breadboards and TI-MSP432 development boards.		
	Thrust Vector Control Intern – Jacobs Space Exploration Group May 2024 to Aug 2024 – Huntsville, AL - Developed thrust vector control testing hardware and software for NASA's Active Inertial Load Simulator at the Marshall Space Flight Center. - Created and ran tests to develop a mathematical model of an electro-mechanical actuator – used Python, MATLAB, and LabVIEW.		
	Control Research Intern – Grenoble Electrical Engineering Lab Jun 2023 to Aug 2023 – Grenoble, FR Implemented inverter control methods for 4-leg (microgrid) topologies and tested them in Simulink and HIL.		
	Engineering Intern (NREIP) – Naval Surface Warfare Center Jun 2022 to Aug 2022 – Bethesda, MD Developed concept hospital sea-train designs, estimated fuel consumption, and electrical power loads.		
Education	M.S. Computer Engineering Virginia Tech May 2025 - Focus: Software and Machine Intelligence - Research Group: COOL Autonomy Lab @ UT Austin		B.S. Electrical & Computer Engineering (Double Major) Virginia Tech May 2024 Focus: Controls, Robotics, and Autonomy
Tools/Libs	ROS2, OMPL, DDS, GNU/Linux, Git, Qt, Simulink, Docker, FreeRTOS, SOLIDWORKS, Autodesk Inventor		
Projects	FPV Wingman ROS2, OMPL, Nav2, C++, Python 3D printed robot arm, built using stepper motors and pulleys - Implemented ROS2 Jazzy control and Gazebo Harmonic simulation		
	Robotic Arm ROS2, OMPL, C++, Python 3D printed robot arm, built using stepper motors and pulleys - Implemented ROS2 Jazzy control and Gazebo Harmonic simulation		